13. RHINOLOGY PART 2

CHOANAL ATRESIA

00:00:01

- Due to persistence of bucconasal membrane
- Associated with CHARGE syndrome
  - C- Coloboma
  - H- Heart defect
  - A - Atresia
  - R - Retardation (growth and mental retardation)
  - G - Genital malformation
  - E - Ear abnormalities
- Can be Unilateral or Bilateral
  - B/l atresia → emergency

Bilateral Choanal Atresia

- Presents as an emergency at birth.
  - Infants are obligate nasal breathers until 3 months of age
  - Choana fails to communicate with the nasopharynx → patient is unable to breathe.
- Cyclical cyanosis: Cyanosis is present at rest and disappears on crying
  - Mechanism: At rest, nose and mouth are blocked → cyanosis
  - On crying, negative intrathoracic pressure opens the mouth → air enters lungs → cyanosis improves
- Crying improves stridor

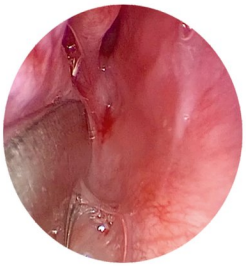
Unilateral Choanal Atresia

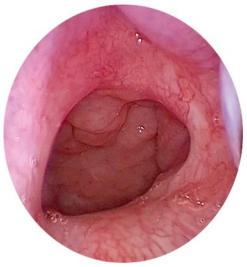
- Not an emergency; presents later in life.
- Triggered during a cold or upper respiratory tract infection (URTI) → difficulty breathing and stridor
- Constant unilateral Rhinorrhea
  - Secretions on the normal side drain into the nasopharynx
  - Secretions on the atresia side cannot drain posteriorly → come anteriorly
  - Results in constant unilateral rhinorrhea, crusting, and skin excoriation
- Crusting, excoriation of skin present

FMGE 2023

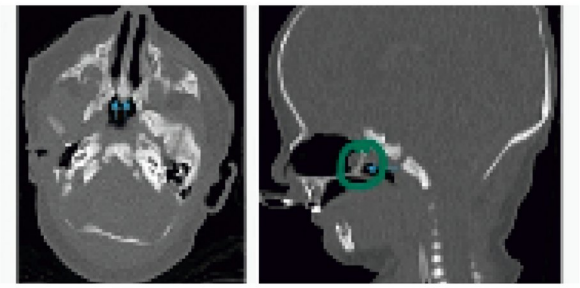
Diagnosis

- Simple bedside test: Pass a Foley's catheter through the nose
  - Normal → Passes into nasopharynx and down to oropharynx
  - Resistance/inability to pass → Indicates choanal atresia.
- Endoscopy examination under anesthesia





- CT scan
  - Confirms diagnosis by showing no communication between the nose and nasopharynx.



Treatment

Bilateral Choanal Atresia:



- Step 1: McGovern's technique
  - Provide a feeding nipple (McGovern's nipple) to allow simultaneous breathing and feeding
  - Restores the airway temporarily.
- Step 2: Tracheostomy to bypass the upper airway and secure it for a longer period.
- Step 3: Definitive repair at 1 to 1.5 years of age.

Unilateral Choanal Atresia:

- Requires no intervention at birth
- Direct surgical repair is performed at 1 to 1.5 years of age

Error Files

- Choanal atresia
  - Cyclical cyanosis → cyanosis that gets better on crying
  - Crying improves stridor
- Crying worsens stridor in
  - Acute epiglottitis
  - Croup
  - Laryngomalacia

SEPTAL DISEASES

00:08:30

Deviated Nasal Septum

NEET PG 2022, FMGE 2023

- Septum does not lie in the midline and fails to divide the nasal cavities into equal halves.
- Extremely common (affects ~95% of patients)

Indiations Of Surgery

- Presence of symptoms, not resolving with medical therapy
- Symptoms include
  - Nasal blockade
  - Headache
  - Sinusitis
  - Smell abnormalities
  - Facial pain
  - External deformities of nose
- Types of Surgery
  - Septoplasty-very conservative type of surgery

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- SMR
- No septal surgery preferred tilL17 years of age

- Cottle's Line
  - Imaginary line drawn from the frontal spine to the anterior nasal spine.
  - Anterior deviations (anterior to line) → Treated only with Septoplasty.
  - Posterior deviations (posterior to line) → Treated with either Septoplasty or SMR

	Septoplasty	SMR
Children	Yes	No
Elevation of flap	single	both
Anterior deviation	Preffered	—
Risk of complication	Very less	High
Risk of recurrence	High	Less

Septal Abscess V/s Septal Hematoma

00:13:30

- Both present with bilateral septal swelling, nasal blockade, pain, and breathing difficulty.
- Septal Hematoma:
  - History of trauma
  - Accumulation of blood between cartilage/bone and overlying perichondrium/mucosa
- Septal Abscess:
  - History of infection or untreated septal hematoma
  - Accumulation of pus between septum and overlying mucosa



	SEPTAL ABSCESS	SEPTAL HEMATOMA
PRESENTATION	Nasal blockade	Nasal blockade
HEADACHE	Yes	No
LYMPHADENOPATHY	Yes	No
FEVER	Yes	NO
EARLY INTERVENTION	• Incision and Drainage ○ Early intervention is strictly required ○ Present in dangerous area of face ○ Infection can travel retrograde into the cavernous sinus → Cavernous sinus thrombosis ○ Abscess → Necrosis of underlying cartilage → Saddle nose deformity	• Incision and drainage + Pressure packing

Septal Perforation

00:16:03

- Hole in the septum allowing communication between left and right nasal cavities
- Cartilaginous portion perforations:
  - Most common site.
  - Causes:

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→ Trauma (finger, nose rings)
→ Inhalation of fumes (asbestos, mustard gas)
→ Cocaine sniffing
→ Diseases (tuberculosis, leprosy, sarcoidosis)

- Bony portion perforations:
  - Typically seen in Syphilis.
- Total septal perforations:
  - Typically seen in Wegener's granulomatosis

Presentation

- Small perforation → Whistling sound during respiration.
- Large perforation → Air causes drying of edges → Crusting → Crust removal leads to bleeding

Treatment

- Most are asymptomatic and require no intervention.
- If symptomatic → Insert a Silastic button

MCQ

Q. A patient presents with chronic sinusitis, nasal crusting, and saddle nose deformity. What is the most likely diagnosis?

FMGE Jan 2026

a. Granulomatosis with polyangiitis
b. Allergic rhinitis
c. Nasal polyposis
d. Rhinoscleroma

Ans: A

Explanation

- Allergic rhinitis & Polyps → Non-erosive, do not cause bone destruction.
- Rhinoscleroma → Causes woody hard granuloma and stenosis, but no saddle nose deformity.
- Granulomatosis with polyangiitis (Wegener's) → Causes total septal perforation, crusting, and bone destruction resulting in a saddle nose deformity

NASAL POLYPS

00:20:07

FMGE 2023

- Definition: Chronically inflamed, edematous tissue that prolapses out.

Antrochoanal Polyp

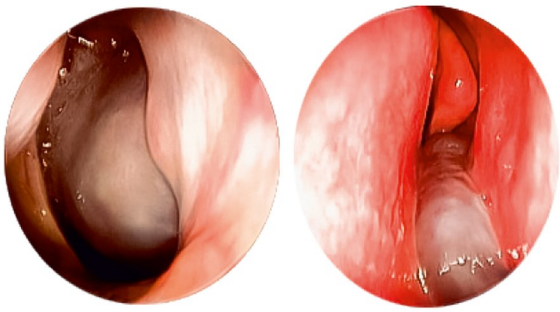
- Origin: Maxillary antrum
- It grows into the nasal cavity reaching nasopharynx
- Single / unilateral mass
- Etiology: Infection (commonly Streptococcus pneumoniae)
- Young Age
- Male predominant

Presentation

- U/l nasal blockade
- Nasal obstruction, discharge
- Widening of nose
- Hyposmia

Diagnosis

- O/e → Greyish white polypoidal mass growing posteriorly



- Next step: CT Scan → Shows unilateral haziness



Treatment

- No role for medical therapy.
- Surgery → FESS (Functional Endoscopic Sinus Surgery).

FESS

- Bulla ethmoidalis lies behind the uncinate process
- Sequence of FESS

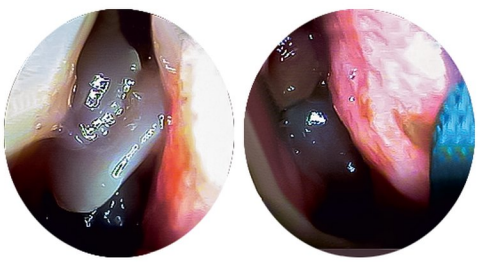
1. Uncinectomy
2. Maxillary sinusotomy
3. Anterior ethmoidectomy
4. Posterior ethmoidectomy
5. Sphenoidotomy
6. Frontal sinusotomy

- FESS has largely replaced the Caldwell-Luc procedure

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Ethmoidal Polyp

- Arises from ethmoidal air cells
- Occur secondary to allergy
- Affect both sides of nose affected
- Multiple masses/polypoidal tissue growing anteriorly to vestibule of nose
- Bilateral



- Bunch of grape appearance on examination
- Associated syndromes:
  - Chronic rhinosinusitis
  - Asthma
  - Cystic fibrosis
  - Kartagener's syndrome: Bronchiectasis, sinusitis, situs inversus.
  - Young syndrome: Sinopulmonary syndrome and azoospermia.
  - Churg-Strauss syndrome:
  - Asthma, fever, eosinophilia, vasculitis,
  - Granuloma
  - Sampter's triad: Nasal polyps, asthma, aspirin intolerance.

Treatment

- Step 1: Medical therapy
  - Steroid → primary
  - Antiallergics
  - Mast cell stabiliser
- If medical therapy fails → CT-scan → FESS

	AC POLYP	ETHMOIDAL POLYP
AGE	Young	Older
ETIOLOGY	Infection(Strep. Pneumonia)	Allergy
NUMBE	Single	Multiple
LATERALLITY	Unilateral	Bilateral
ORIGIN	Maxillary Sinus	Ethmodial Air Cells
GROWTH	Posteriorly	Anteriorly
MEDICAL TREATMENT	No Role	Yes
SURGERY	FESS	FESS
RECURRENCE	Low	High

MCQ

Q. A 54-year-old male presented with nasal obstruction. On examination there was an irregular mass protruding from the choanae. What is the possible diagnosis?

a. Thornwaldt cyst
b. Antro-choanal polyp
c. Hypertrophic rhinitis
d. Adenoid hypertrophy

Ans: B

YOURWISH

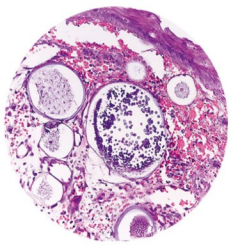
GRANULOMATOUS DISEASES OF THE NOSE

00:34:08

- Rhinosporisiosis
- Rhinoscleroma

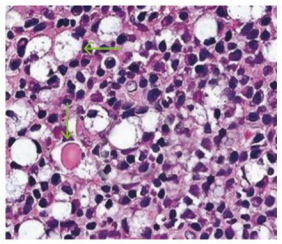
Rhinosporidiosis

NEET PG 2018

- Caused by protozoa→ rhinosporidium seeberi
  - Aquatic protozoa
  - Endemic in South India.
  - Transmission: Waterborne route (contact with contaminated ponds)
- Presentation: Nasal blockade and bleeding
- O/e
  - Red mass with whitish dot on surface →strawberry mass
- CT scan can be done
- Treatment: Cauterize the base + excision of mass
- HPE: Shows multiple sporangia with spores inside

RHINOSCLEROMA

- Granulomatous condition by bacteria → Klebsiella rhinoscleromatous/frisch bacillus
  - Endemic in North India.
  - Transmission: Air droplets.
- Stages of Disease:

1. Atrophic Stage: Bacteria destroys mucin-producing cells → Nose becomes dry → Crusting and foul smell
2. Granulomatous Stage: Immune response creates a central necrotic area surrounded by inflammatory cells → "Woody hard granuloma" → Results in a "Hebra nose" deformity
3. Stage of Cicatrization: Healing via fibrosis → Stenosis/narrowing of the nose → Results in a "Tapir nose" deformity

- Histopathology: Presence of Mikulicz cells and Russell bodies
- Culture on mcckoney agar
- Management:
  - Medical therapy: Streptomycin, Tetracycline, and Rifampicin for 4-6 weeks
  - Therapy is stopped only after two consecutive cultures (4 weeks apart) are negative

Feature	Rhinosporidiosis	Rhinoscleroma
Causative organism	Rhinosporidium seeberi (aquatic protozoan-like organism)	Klebsiella rhinoscleromatis
Mode of transmission	Water exposure	Air droplets
Common region	South India	North India
Symptoms	Nasal blockage + bleeding	Crusting, foul smell Granuloma Narrowing / stenosis of nose
Characteristic appearance	Strawberry-like mass	
Nasal deformity	—	Hebra nose, Tapir nose

YOURWISH

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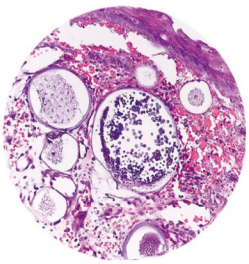
Diagnosis	• HPE-final diagnosis • Excision and cauterization of base • Dapsone to prevent recurrence	• Culture ○ Positive on MacConkey agar • Mikulicz cells, Russell bodies on HPE
Treatment	Excision and cauterization of base	• Treated with tetracyclin, streptomycin and rifampicin • Debridement of mass if needed

MCQ

Q. South Indian male farmer presents to the ENT OPD with complaints of reddish mass coming out from the nose and histopathology image as given below. What is the likely diagnosis?

a. Rhinosporidiosis
b. Rhinoscleroma
c. Rhinophyma
d. Nasal polyp





Ans: A

RHINITIS

00:45:22

ALLERGIC RHINITIS	NON ALLERGIC RHINITIS/VASOMOTOR RHINITIS
• Serum IgE elevate • Presentation ○ Sneezing ○ Itching ○ Watery discharge ○ Nasal blockade • Mucosa is Pale, Boggy, edematous • Management ○ First line: Intranasal corticosteroids + Antihistamines + Decongestants • If unresolved: ○ Skin prick test / Radioallergosorbent test to find specific allergen → Immunotherapy (Subcutaneous or Sublingual) ○ Immunotherapy types: SCIT, SLIT • Specific Signs ○ Allergic shiners: Dark circles around the eyes. ○ Allergic salute: Constant upward rubbing of the nose. ○ Darier's line: Dark transverse crease on the nasal dorsum.	• Serum IgE normal • Presentation ○ Sneezing ○ Itching ○ Watery discharge ○ Nasal blockade • Pathophysiology: ○ Imbalance of autonomic nervous system (Parasympathetic hyperactivity + Sympathetic hypoactivity) • O/e ○ Mulberry turbinate or hypertrophic turbinate • Management: ○ 1st Line: Intranasal corticosteroids + Antihistamines+ Decongestants ○ If medical therapy fails → Surgical reduction of inferior turbinates ○ Final surgical option → Vidian neurectomy.

MCQ

Q. A patient was presented to the OPD with complaints of greenish black matter in the nose with merciful smelling. What is the diagnosis?

a. Allergic rhinitis
b. Atrophic rhinitis
c. Rhinitis caseosa
d. Rhinitis sicca

Ans: B

- Atrophic Rhinitis:
  - Atrophy of entire sinus mucosa, nasal mucosal glands, and olfactory epithelium.
  - Results in a dry nose, greenish foul-smelling crusts, and merciful anosmia.
  - Merciful Anosmia: Patient does not perceive the foul smell due to olfactory epithelium atrophy, but people around them do.

Other Rhinitis Conditions

- Rhinitis sicca
  - Similar to atrophic rhinitis but limited only to the anterior 1/3rd of the nose.
  - Etiology: Constant exposure to dry air currents, hot/humid environments.
  - Symptoms: Nasal crusting, foul smell, nasal blockade.
  - Treatment: Saline irrigation spray and changing profession
- Rhinitis caseosa
  - Accumulation of cheesy material within sinuses.
  - Etiology: Chronic obstruction of nose/sinuses (e.g., polyp, foreign body, DNS) → retention of secretions
  - Symptoms: Cheesy, thick, mucoid, foul-smelling discharge with crusting
  - Treatment: Surgical removal of the obstruction to clear secretions and restore patent airway
- Tuberculosis
  - Primary TB: Occurs directly inside the nose.
  - Secondary TB: Occurs secondary to an active pulmonary TB focus.
  - Presentation: Granuloma, usually affecting the cartilaginous portion of the nose
  - Complication: Septal perforation.
  - Diagnosis: GeneXpert or growth on LJ (Lowenstein-Jensen) medium.
  - Treatment: Anti-tubercular therapy (ATT).
- Lupus vulgaris
  - Low-grade, indolent tubercular infection of the nose
  - Usually not associated with active pulmonary TB
  - Specific Sign: Apple jelly nodule
    → Nodule blanches upon application of pressure.
  - Treatment: ATT.
- Sarcoidosis
  - Non-caseating granuloma inside the nose or sinuses.
  - Specific Sign: "Strawberry mucosa"
    → Differential Note: "Strawberry polyp" indicates Rhinosporidiosis.
  - Complications: Destructive lesions affecting the cartilaginous septum → Saddle nose deformity, chronic sinusitis
  - Can be associated with systemic sarcoidosis manifestations.

Atrophic Rhinitis

00:56:22

- Atrophy of all nasal structures (mucosa, sinus mucosa, olfactory epithelium, mucus-producing cells)
- Symptoms: Dry nose, crusting, foul smell
- Examination Findings:
  - Dry crusting visible
  - Nasal cavity appears wide and roomy due to shrinkage
- Paradoxical symptom: Patient complains of nasal blockade despite wide roomy chambers (loss of sensation of airflow)
- Merciful anosmia present
- Causes (Mnemonic: HENIA + R):
  - H → Hereditary
  - E → Endocrinological problems
  - R → Racial
  - N → Nutritional deficiencies (Vitamin A, D, E)
  - I → Infective causes (Klebsiella ozaenae, most common cause, also called Perez bacillus)
  - A → Autoimmune causes
- Treatment:
  - Step 1: Medical therapy.
    → Irrigate nose to keep it clean and moist.
    → Solution: Alkaline nasal douching solution.
    → Composition (mixed in 280 ml water): 2 parts sodium chloride + 1 part sodium bicarbonate + 1 part sodium biborate.
    → Other options: Placental extracts, estrogen/estradiol spray, Kemecetine anti-ozaena solutions
  - Step 2: Surgery (If medical therapy fails)
    → Young's operation: Complete surgical closure of the nasal cavity (elevating mucosa on septum, lateral wall, and floor).
  - Modified Young's operation: Leaves a 10 mm gap for breathing
  - Lautenschlager's operation: Medialization of the lateral wall of the nose.

EPISTAXIS

01:01:49

- Any bleeding from nose

Blood Supply Of Nasal Septum

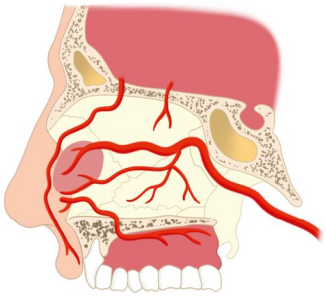
- Upper half: Internal Carotid Artery (ICA) circulation.
  - ICA→ Ophthalmic artery.
  - Ophthalmic artery → Anterior ethmoidal artery (supplies anterosuperior septum) + Posterior ethmoidal artery (supplies posterosuperior septum).
- Lower half: External Carotid Artery (ECA) circulation.
  - ECA → Facial artery + Internal maxillary artery.
  - Facial artery → Superior labial artery.
  - Internal maxillary artery → Greater palatine artery + Sphenopalatine artery (both supply posteroinferior septum)
- Vascular Plexuses:
  - Little's area / Kiesselbach's plexus:

YOURWISH

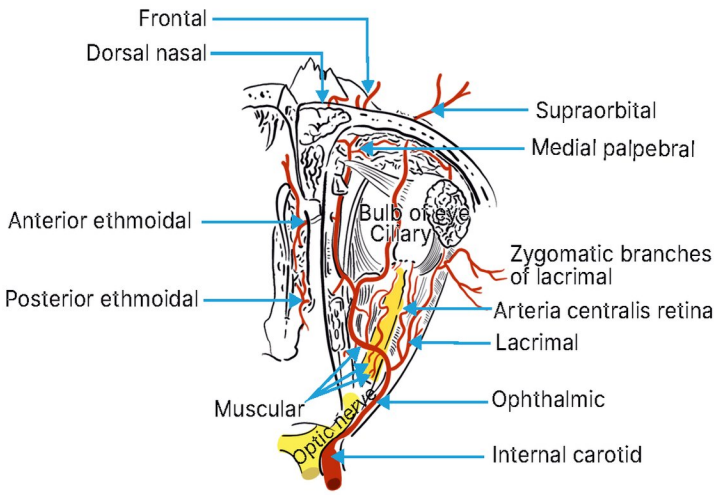
→ Anastomosis at the anteroinferior part of the nose
→ Any bleed → Anterior epistaxis

- Woodruff's plexus:
  → Venous plexus at the posterior end of the inferior turbinate
  → Any bleed → Posterior epistaxis

- Artery of epistaxis: Sphenopalatine artery.



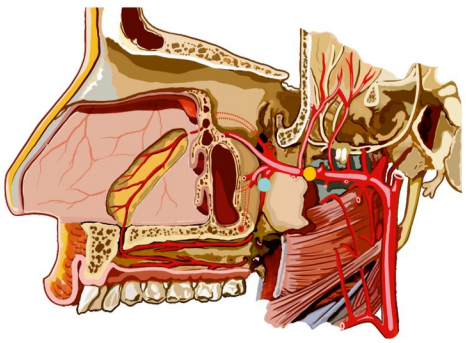
Course Of Anterior And Posterior Ethmoidial Artery



- Internal carotid artery gives a branch → ophthalmic artery
- Ophthalmic artery crosses optic nerve (lateral to medial) and enters medial wall of orbit
- Divides into Anterior and Posterior ethmoidal arteries along the medial wall
- Rule of 24-12-6 (Using anterior lacrimal crest as landmark):
  - Anterior ethmoidal artery: 24 mm behind crest.
  - Posterior ethmoidal artery: 12 mm behind anterior ethmoidal artery.
  - Optic canal: 6 mm behind posterior ethmoidal artery.
- Anterior ethmoidal artery passes through anterior ethmoidal foramen and ethmoidal air cells to supply septum
  - Surgical Risk: Can be traumatized during ethmoidectomy

Course Of Sphenoplatine And Greater Palatine

- Internal Maxillary Artery Branches:
  - Runs within the pterygopalatine fossa (behind posterior wall of maxillary sinus).
- Ligation of the internal maxillary artery is done in the pterygopalatine fossa.
- Sphenopalatine artery: Ligated ~8 to 10 mm behind the middle turbinate.
- Greater palatine artery: Dips down to palate, ligated near the 3rd molar



Causes Of Epistaxis

- Local Causes: Trauma, tumor, granulomatous diseases, rhinosporidiosis
- Systemic Causes:
  - Hypertension (most common systemic cause).

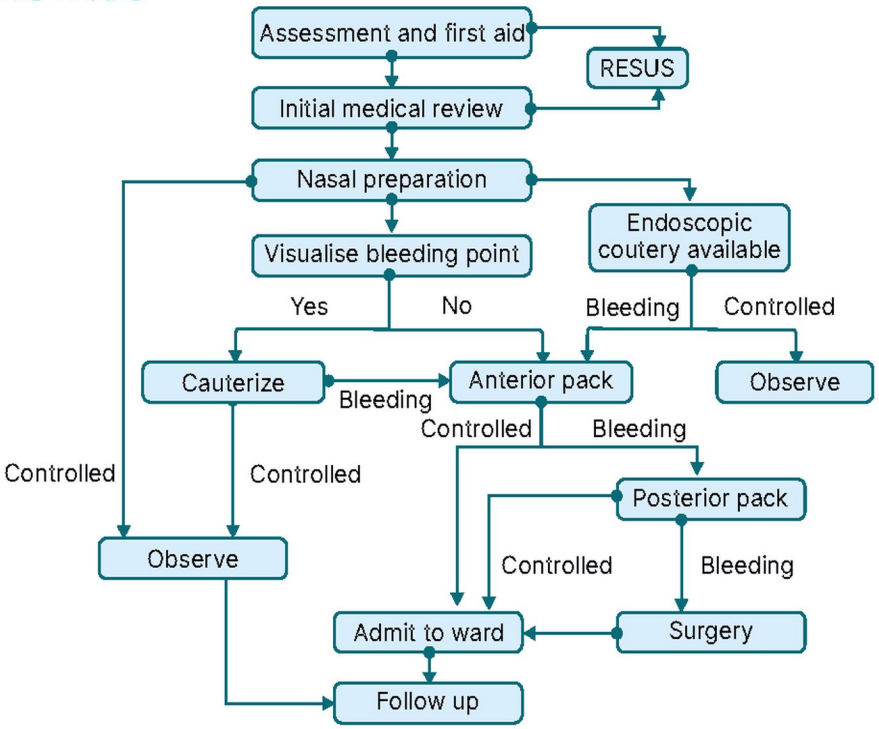
YOURWISH

- Blood disorders: Hemophilia, von Willebrand's disease, sickle cell anemia.
- Coagulation/Bleeding disorders.
- Blood vessel disorders: Hereditary hemorrhagic telangiectasia.
- Medications: Oral anticoagulants, thrombolytics, antiplatelets, OCPs
- Liver disease (e.g., cirrhosis due to alcoholism or drugs)
- Renal disease

	ANTERIOR EPISTAXIS	POSTERIOR EPISTAXIS
Incidence	More common	Less common
Site	Littles area	Woodruff plexus/ SPA
Age	children	Ager 40 years of age
Cause	Trauma	HTN
Bleeding	Mild can be controlled by local pressure of anterior pack	Bleeding is severe,requires hospitalization and postnasal pack

PROTOCOL FOR EPISTAXIS

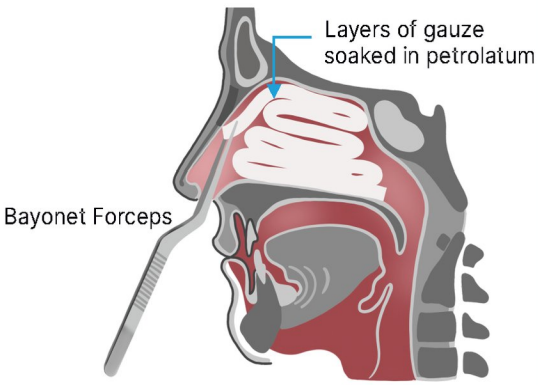
01:10:20



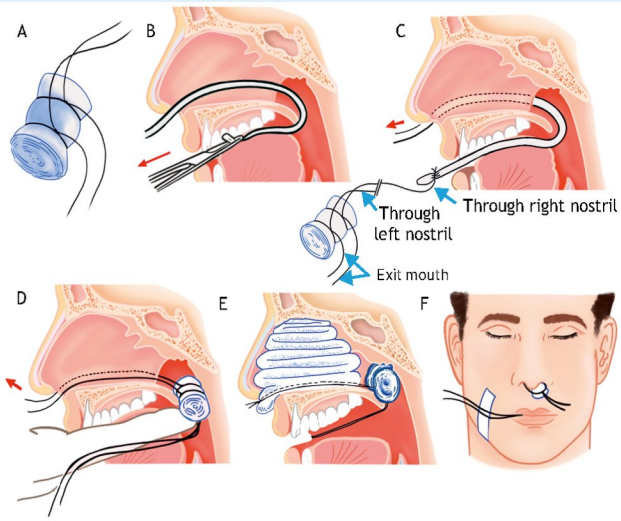
- Step 1: Trotter's Method.
  - Pinch nose and bend forward for ~15 minutes
- Step 2: Initial medical review and resuscitation.
- Step 3: Nasal preparation for Endoscopy.
  - Spray Xylocaine (numbing) and Adrenaline (reduces bleeding)
- Step 4: Endoscopic Cauterization.
  - If a single bleeding point is seen → Use chemical cautery (Silver nitrate solution) or electrical cautery (monopolar/ bipolar).
- Step 5: Anterior Nasal Packing.
  - Done if no bleeding point is seen, endoscopy is unavailable, or cautery fails.

- Step 6: Posterior Nasal Packing.
  - Done if anterior packing fails.
- Step 7: Surgical Ligation of Artery (Last resort).
  - Order of ligation:

1. Sphenopalatine artery
2. Internal maxillary artery
3. External carotid artery (if bleeding is from lower half)
4. Ethmoidal artery (if bleeding is from upper half)



Anterior Nasal packing



Posterior Nasal Packing

- Procedure: Pass Ryle's/infant feeding tube through nose to mouth → attach threaded ribbon gauze → retract tube from nose → gauze sits in nasopharynx → then block anterior nose.

MCQ

Q. Ligation of artery

a. Sphenopalatine A
b. Internal maxillary artery
c. Ethmoidal artery
d. External carotid artery

Ans: Sphenopalatine A → Internal maxillary artery → Ethmoidal artery or External carotid artery

One Liners

- Most common cause of epistaxis is Trauma
- Most common cause of epistaxis in children is finger trauma
- Most common cause of epistaxis with foul smelling nasal discharge which is unilateral in children is → foreign body
- Most common cause of recurrent unprovoked profuse epistaxis in adolescent male is JNA
- Most common cause of epistaxis in elderly is HTN
- Most common site of epistaxis is from Sphenopalatine Artery
- Anterior epistaxis if venous occurs from the Reterocolumella vein
- The artery of epistaxis is Sphenopalatine Artery
- Posterior epistaxis occurs from Woodruff plexus or Sphenopalatine artery
- TESPAL stands for Transnasal Endoscopic sphenopalatine artery ligation

MCQ

Q. A 70 year old patient with epistaxis, patient is hypertensive with BP of 200/100 mm hg. On examination no active bleeding is noted next step in the management is

a Observation
b. Internal maxillary artery ligation
c. Anterior and posterior nasal packing
d. Anterior nasal packing

Ans: A

Q. Source of epistaxis after ligation of external carotid artery is

a Maxillary artery
b. Greater Palatine artery
c. Superior labial artery
d. Ethmoidal Artery

Ans: D

Q. In case of uncontrolled epistaxis ligation of internal maxillary artery is done in the

a. Maxillary antrum
b. Pterygopalatine fossa
c. At the neck
d. Medial wall of orbit

Ans: B

Q. Initial treatment of severe epistaxis

a. The patient should be placed head down supine with a pillow under the shoulders.
b. Trotter's method includes encouraging the patient to swallow the blood, to allay psychological stress and thereby reduce blood pressure.
c. A postnasal pack should be considered when the haemoglobin falls below 10 /litre.
d. Epistaxis balloons should be inflated with sterile water.
e. A central venous pressure line is mandatory

Ans: C

FRACTURES OF FACE

01:20:58

- Middle third and lower thyroid mostly
- Nasal fracture (M/c)> zygomatic fractures> maxillary fractures> floor of orbit fractures

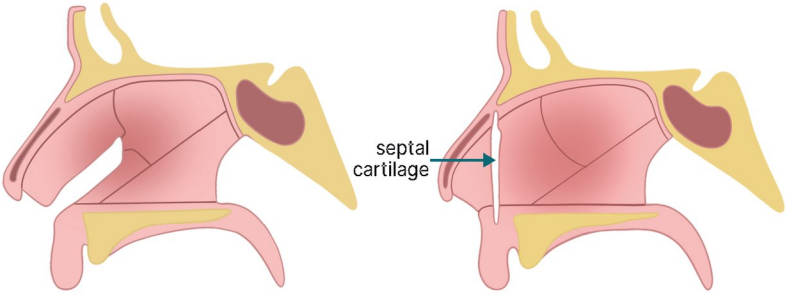
Nasal Fractures

- Class 1-Chevallet fracture
  - Vertical fracture line
  - Direction of injury from below

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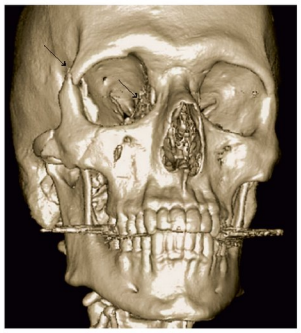
- Class 2- Jarjaway fracture
  - Horizontal fracture line
  - Direction of injury from front
- Class 3- Naso orbital ethmoid fracture
  - Entire component gets crushed rotated nasal upwards → pig nose deformity
  - High risk of CSF leak

FMGE 2022, 2025 INICET 2024, 2025



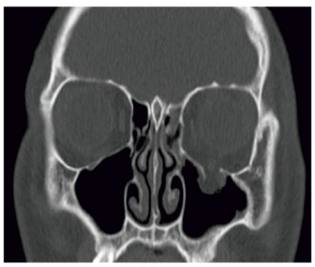
Zygomatic Fracture'

- Three fracture line
  - Zygomaticomaxillary suture
  - Zygomaticofrontal suture
  - Zygomaticotemporal suture
- Tripod fracture
- Crush of zygoma → step deformity



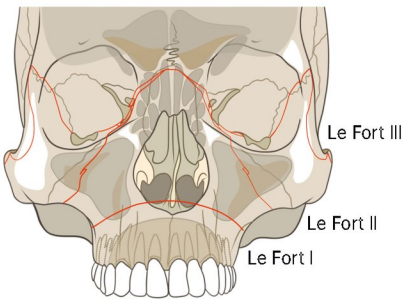
Floor Of Orbit Fracture

- Floor of orbit has no support inferiorly because of the maxillary sinus.
- Called Blow-out fracture
- Orbital contents herniate into maxillary sinus → tear drop sign
- Inferior rectus and inferior oblique gets entrapped → diplopia



Maxillary Fracture

- A/k/a Lefort fracture

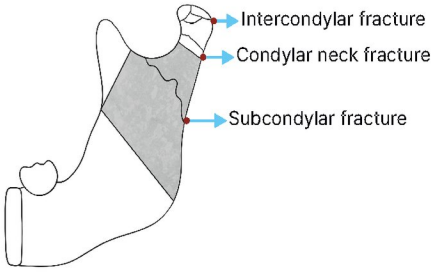


Le forte 1	• Fracture through floor of nose and floor of maxilla • Aka Transverse fracture / Guerin's fracture • Greater palatine artery rupture → bluish discolouration of palate called geurin's sign • Hanging palate and hanging teeth seen
Le Forte 2	• Pyramidal or pyramid-shaped fracture. • Course of fracture line: Starts at the Nasion → Lacrimal bone → Medial wall of the orbit → Floor of the orbit → Maxilla. • Key characteristic feature: Hanging maxilla
Le Forte 3	• Course of fracture line: Nasion → Lacrimal bone → Medial wall of the orbit → Floor of the orbit → Lateral wall of the orbit → Zygoma. • Key characteristic feature: Disconnects the cranial cavity from the facial skeleton • Aka Craniofacial disjunction • Associated with a high risk of CSF leak (CSF rhinorrhea)

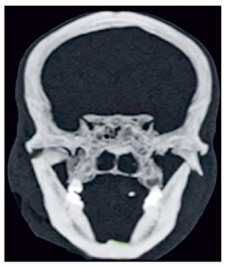
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Fracture Of Mandible



- M/c site: Subcondylar region
  - Fractures rarely occur at the direct site of trauma
  - Impact forces are transmitted to the opposite side (counter-coup effect) → resulting in an opposite subcondylar fracture.
- Guardsman fracture / Parade ground fracture
  - Patient falls forward directly on the symphysis menti (common in guards/marching personnel)
  - Fracture at the symphysis menti + Bilateral subcondylar fractures
- Management of Mandibular Fractures
  - Treated via closed reduction with internal fixation OR open reduction with internal fixation.
  - Do NOT immobilize the mandible for more than 3 to 4 weeks.
  - Prolonged immobilization (>4 weeks) → Leads to permanent TMJ ankylosis and inability to open the mouth



One Liners

- Most common site of mandibular fractures is at Subcondylar region
- Subcondylar fractures occur due to trauma to body of mandible from opposite side
- When a closed reduction is done, mandible should not be immobilized for 3-4 weeks.
- Other name for guardsman fracture Parade ground fracture

MCQ

Q. Most important primary care in faciomaxillary trauma is to

a. Assess level of consciousness
b. Ensure adequate airway
c. Look for CNS injury
d. Immediate fracture reduction

Ans: B

Q. Ideal time for correction of fracture nasal bone is

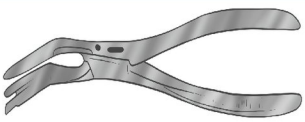
a. Immediately
b. After 1 week
c. After 3 weeks
d. After 6 weeks

Ans: B

Forceps

ASH SEPTUM FORCEPS

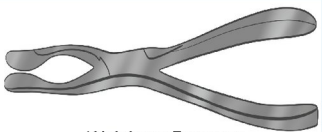
- Has angle
- Used for Nasal septum



Asch's Forceps

WALSHAM SEPTUM FORCEPS

- Used for Nasal bone reduction



Walsham Forceps

MCQ

Q. Tripod fracture is seen in
a. Mandible
b. Maxilla
c. Nasal bone
d. Zygoma

Ans: D

Q. Nerve to be traumatized in zygomatic fractures are
a. Facial nerve
b. Zygomaticotemporal nerve
c. Supraorbital nerve
d. Infraorbital nerve

Ans: D

CSF RHINORRHEA

01:34:40

- Leakage of cerebrospinal fluid into the nasal cavity.
- M/c cause: Trauma (95%)
- Rare cause:
  - ↑ Intracranial pressure (e.g., Benign Intracranial Hypertension) → Thinning of the bone separating the cranial and nasal cavities → Leakage
- Clinical presentation:
  - History of trauma followed by watery nasal discharge

Diagnosis

- Clinical Bedside Tests (To differentiate CSF from nasal mucin):
  - Sniff Test: Nasal fluid can be sniffed back; CSF cannot be sniffed back.
  - Handkerchief Test: Nasal fluid stiffens the cloth on drying; CSF does not stiffen the cloth
  - Reservoir Sign: Bending forward in a gravity-dependent position → ↑ Watery discharge
  - Halo Test / Double Ring Sign / Target Sign (For traumatic cases with mixed blood): Filter paper showing a central red dot surrounded by a clear peripheral ring
  - Valsalva Maneuver / Coughing: Increases intracranial pressure → ↑ CSF flow.
  - Note: All clinical tests are non-specific
- Confirmatory Laboratory Investigations:
  - IOC: Beta-2 Transferrin test (Protein specific to CSF, aqueous humor, and perilymph)
  - Newer highly selective test: Beta-trace protein (Specific only to CSF)

Radiological Investigations

- HRCT: Used to identify the exact site of the bony defect
- CT Cisternography: Used to identify active leaks
- MRI: Used to compare fluid intensity (CSF vs. nasal fluid)

Management:

- Transnasal Endoscopic Repair of the defect.

MCQ

Q. Target sign is suggestive of

a. Traumatic CSF rhinorrhea

b. Spontaneous csf rhinorrhea

c. Angiofibroma

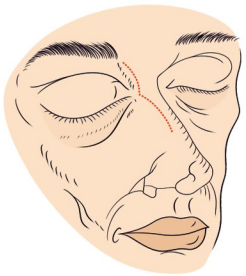
d. Glomus tympanicum

Ans: A

TUMOURS OF NOSE AND PNS

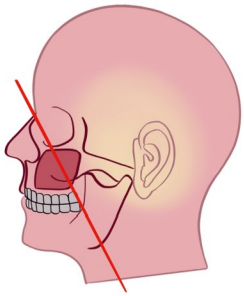
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Osteoma

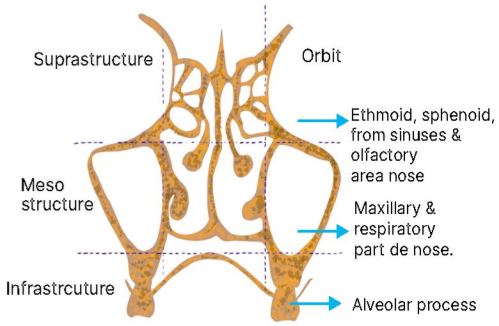


- MC benign tumour of PNS
- Site: frontal > frontoethmoid region
- HPE: Dense lamellar bone lacking Haversian canals.
- Associated Condition: Gardner Syndrome (Osteomas + Colorectal polyps).
- Management:
  - Lynch-Howarth external frontoethmoidectomy
    → The Lynch-Howarth incision starts at the medial edge of the eyebrow and curves down midway between the nasal dorsum and medial canthus and extended downwards to access frontal and ethmoid sinus

Sino-nasal Malignancies



- Etiology: Exposure to carcinogens (smoke, alcohol, tobacco).
- M/c histopathological type: Squamous Cell Carcinoma.
- Woodworkers inhaling wood dust are at high risk for Adenocarcinoma.
- Prognostic Classifications:
  - Ohngren's Line:
    → Imaginary line from the medial canthus to the angle of the mandible.
    → Tumors below the line → Good prognosis
    → Tumors above the line → Poor prognosis
  - Lederman's Classification:
    → 2 imaginary line from medial wall of orbit and 2 from floor of maxillary sinus
    → Higher the tumor location → Poorer the prognosis.
    → Lower the tumor location → Better the prognosis



- Presentation
  - Dull pain over maxillary sinus
  - Ipsilateral epiphora
  - Blood stained nasal discharge
  - Nasal obstruction
  - Toothache
- Diagnosis:
  - Investigation of choice: Biopsy
  - Radiological studies are done prior to Biopsy.
- Management
  - Surgical Management:

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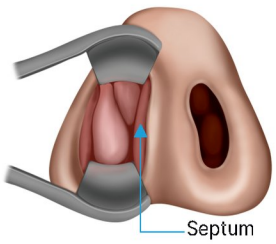
→ Total Maxillectomy is performed to achieve negative margins.
→ Structures removed: Maxilla, ipsilateral palate, ipsilateral eyeball, lateral wall of nose, ethmoidal air cells.
→ Incision used: Weber-Fergusson incision.
- Course of Weber-Fergusson: Lower eyelid → Nasofacial sulcus → Winds around the nasal ala → Crosses philtrum to the upper lip

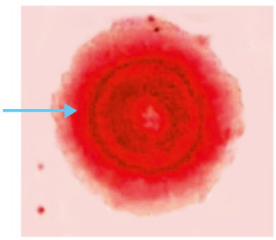
IMAGE BASED QUESTION

01:48:26

ANTERIOR RHINOSCOPY

- Only the inferior and middle turbinates are visible
- The Superior turbinate cannot be seen on anterior rhinoscopy
- Medial wall is septum
- Instruments for Rhinoscopy
  - Thudicum's or Killian's nasal speculum → Used for Anterior Rhinoscopy
  - St. Clair Thomson mirror → Used for Posterior Rhinoscopy
- Endoscopic Nasal View
- Superior turbinate is only visible endoscopically after decongestion and xylocaine application

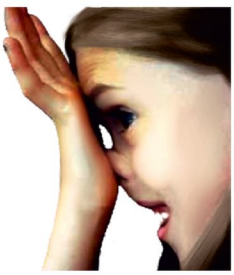




ALLERGIC RHINITIS

- Dark circles around the eyes → Allergic shiners
- Repeated upward rubbing of the nose → Allergic salute
- Transverse dark line on the dorsum of the nose → Allergic crease or Darriers line

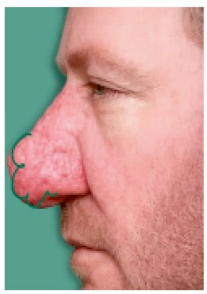






RHINOPHYEMA

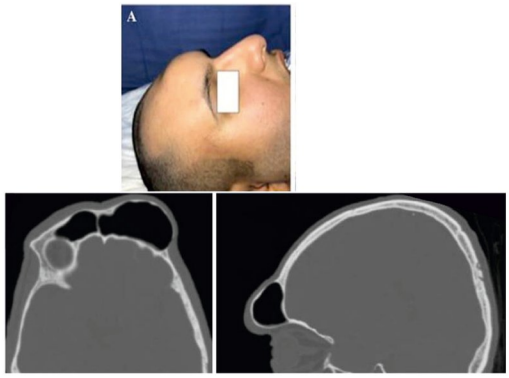
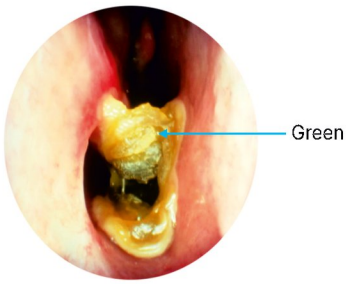
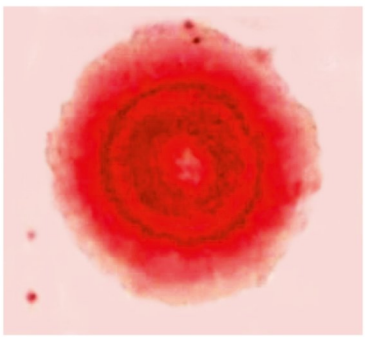
- Hypertrophy of sebaceous glands
- Treatment: Laser debridement with skin grafting



ANTERIOR DEVIATION OF NASLA SEPTUM



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PNEUMATOCELE	• Air filled frontal sinus 
ORBITAL CELLULITIS	• Secondary to ethmoid sinus infection 
RHINITIS SICCA & ATROPHIC RHINITIS	• Green crust present seen in ○ Rhinitis sicca → anterior 1/3rd of nose ○ Atrophic rhinitis → entire nose 
TARGET SIGN	

PYQ

01:54:50

Q. A 40-year-old patient presents with recurrent and severe nosebleeds from the anterior nasal septum. The bleeding has been refractory to nasal packing and chemical cautery. A decision is made to proceed with surgical ligation to control the bleeding. Which of the following arteries is the primary target for ligation in the management of this patient's anterior epistaxis?

a. Sphenopalatine artery
b. Anterior ethmoidal artery
c. Facial artery
d. Internal maxillary artery

Ans: A

Q. Identify the fracture given in diagram:



a. Chevallet fracture
b. Jarjaway fracture
c. Le fort 1
d. Simple fracture

Ans: A

Q. A 45-year-old male was admitted with respiratory distress. CT showed a nasal polyp with fluid collection in the sinus. Drainage of which of the following is obstructed?

a. Ethmoidal sinus
b. Sphenoidal sinus
c. Frontal sinus
d. Maxillary sinus

Ans: D

Q. What is the preferred initial management for a patient presenting with cerebrospinal fluid (CSF) rhinorrhea?



a. Functional Endoscopic Sinus Surgery (FESS)
b. Administration of steroids
c. Bed rest with head elevation and avoidance of straining activities
d. Immediate surgical repair without initial medical management

Ans: C

Q. Which structure does the anterior ethmoidal sinus drain into?

a. Middle meatus
b. Inferior meatus
c. Superior meatus
d. Sphenoethmoidal recess

Ans: A

14. SINUSITIS AND COMPLICATIONS

SINUSITIS

00:00:10

FMGE July 2023
INI CET July 2021, May 2023

- Inflammation of mucosa of paranasal sinuses is called sinusitis
- Duration less than 3 weeks → Acute sinusitis
- Duration 3 to 12 weeks → Subacute sinusitis
- Duration more than 12 weeks → Chronic sinusitis
- Recurrent sinusitis
  - 4 or more episodes per year
  - Each episode lasts 7 to 10 days
- Most common sinus affected in adults: Maxillary sinus
- Most common sinus affected in children: Ethmoid
- Least common sinus affected: Sphenoid sinus

Etiology

- Exciting factors :
  - Nasal infection: Most acute sinusitis starts as viral infections followed by bacterial invasion
  - Swimming & driving
  - Trauma
  - Dental infection (Especially because maxillary sinus is close to first molar and second molar)
- Predisposing factors :
  - Obstruction to ventilation of sinuses (Polyp, Edema, Rhinitis)
  - Stasis of secretion in nasal cavity (cystic fibrosis)
  - Allergy
- Systemic factors:
  - Attack of exanthematous fever (measles,chicken pox,whooping cough)
  - Nutritional causes
  - Immune deficiencies

Bacteriology

- Bacteria:
  - Streptococcus pneumoniae (M/c organism)
  - H. influenza
  - Moraxella
  - Streptococcus pyogenes
  - Staph aureus
- Anerobic organism and mixed infections are seen in sinusitis of dental origin

Clinical Features Of Acute Sinisitis

Major	Minor
• Nasal obstruction	• Headache

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• Facial pain or pressure • Purulent nasal discharge • Fever • Nasal congestion • Hyposmia or anosmia • Facial congestion or fullness	• Headache • Cough • Halitosis • Ear pain and pressure • Dental pain

- For diagnosis :2 major symptoms OR 1 major + 2 minor symptoms

	Maxillary	Frontal	Ethmoid	Sphenoid
Headache	Usually seen over cheeks and forehead and can be confused with frontal sinusitis	Localised to frontal sinuses above medial canthus	Localised over bridge of nose, medial and deep to the eye	Occiput or vertex
Tenderness	Cheek	Over frontal sinus	Bridge of nose	Nil
Redness and oedema	Cheek	Upper eyelid puffiness	Both eyelids become puffy	Nil
Site of nasal discharge	Middle meatus	Vertical streak of mucopus high up in anterior portion of middle meatus	Seen in middle or superior meatus depending on involvement of anterior or posterior group	Seen on posterior rhinoscopy on the roof and posterior wall of nasopharynx
Early complications	Orbital cellulitis, Osteomyelitis, Frontal sinusitis	Orbital cellulitis, Osteomyelitis, intracranial abscess	Orbital cellulitis, vision loss, cavernous sinus thrombosis, intracranial abscess	

Diagnosis

Acute

- Clinical
- Diagnostic nasal endoscopy
- X-ray is first investigation that is done
- CT scan
  - No role in uncomplicated acute sinusitis
  - Done only if complication is suspected
- Gold standard for diagnosis: Sinus puncture and bacterial culture
- Accurate method of diagnosis is sinuscopy and biopsy

Chronic Sinusitis

- NCCT of Nose and Paranasal sinus is done for diagnosis

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Treatment

Acute Sinusitis

- Mainly medical treatment: Antibiotics + decongestants + analgesics + Hot fomentation
- Surgery
  - If no response to medical treatment
  - If it is progressing to complication
  - FESS (Functional endoscopic sinus surgery)/ Ethmoidectomy/ Trephination of frontal sinus

Chronic Sinusitis

FMGE July 2023

- Maximum medical therapy: Antibiotics with antral irrigation is tried and Intranasal corticosteroids for 3 months along with Nasal wash
- If symptoms persist after this → FESS

Fess

- First open the maxillary sinus
- Uncinate process → Boomerang-shaped bone in anterior part of nose
- Behind uncinate process is Bulla ethmoidalis
- Space between uncinate and bulla → Hiatus semilunaris / infundibulum
- Opening of maxillary sinus is in Infundibulum
- Steps

Uncinectomy
↓
Middle meatal antrostomy
↓
Remove anterior ethmoidal cells
↓
Remove posterior ethmoidal cells
↓
Sphenoidotomy
↓
Frontal sinusotomy

INI CET July 2021
00:10:44

Fungal Sinusitis

NON-INVASIVE FUNGAL SINUSITIS	INVASIVE FUNGAL SINUSITIS
• Occurs in immunocompetent patients • Caused by Aspergillus and dematiaceous species	• Occurs in immunocompromised patients • Caused by: ○ Aspergillus fumigatus ○ Rhizopus, Mucor
• Presents as: ○ Fungal ball ○ Allergic fungal rhinosinusitis	• Presents as: ○ Invasive aspergillosis ○ Mucormycosis • Present chronically as Chronic invasive sinusitis
• Antifungals usually not required	• Antifungals are required along with radical debridement

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Bent And Kuhn Criteria For Afrs

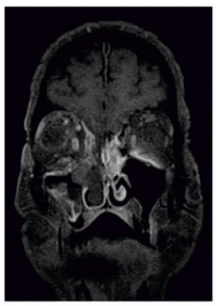
Major	Minor
• ↑ IgE → Type I hypersensitivity • Nasal polyposis • Characteristic CT findings (Double density) • Eosinophilic mucin without invasion • Positive fungal stain	• Asthma • Unilateral disease • Bone erosion • Fungal cultures • Charcot-Leyden crystals • Serum eosinophilia



Mucormycosis

- Black turbinate sign → seen in mucormycosis
- Caused by Mucoraceae family
- Spread is via blood vessels / vascular invasion
  - Black turbinate sign on MRI
  - Black eschar on clinical examination
- Treatment
  - Radical debridement of all dead tissue f/b liposomal amphotericin B

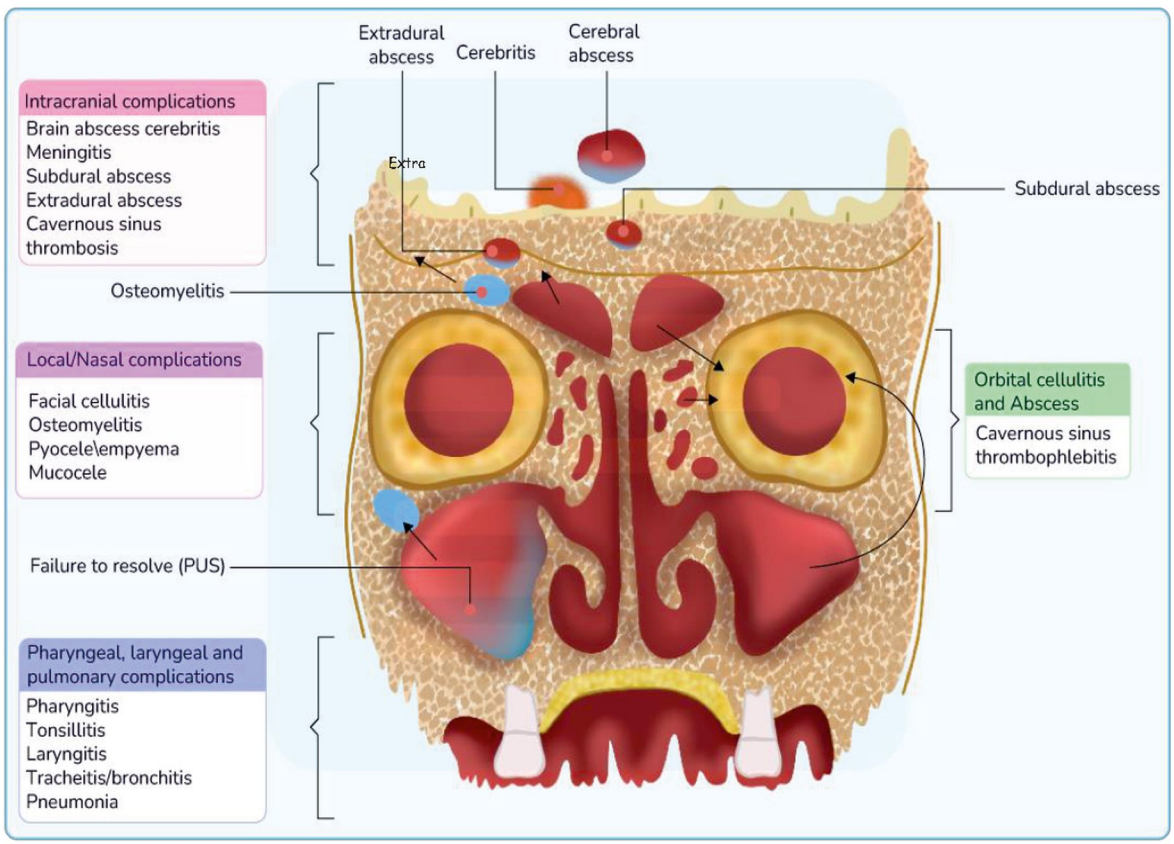




Complications Of Sinusitis

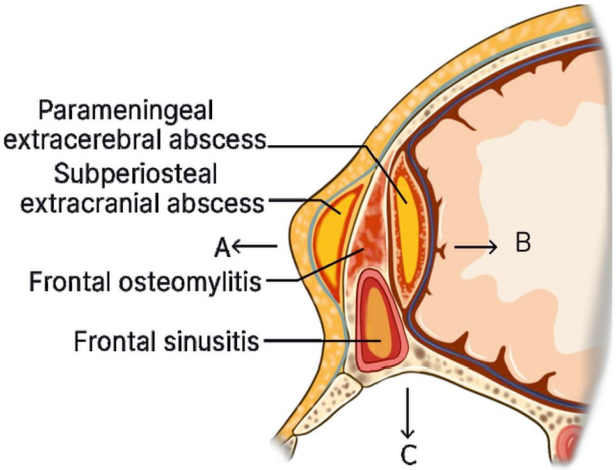
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- Divided into
  - Local complications
  - Orbital complications
  - Intracranial complications



Local Complications

- Retention cyst
  - Cystic transformation of mucosa
- Mucocele
  - Due to blockage of ostium
  - Retention of secretion in an epithelial-lined sac
- Pyocele
  - Infected mucocele
- Orbital osteomyelitis
  - If infection spreads to adjacent bone / orbit
- Frontal bone osteomyelitis → Pott's puffy tumor

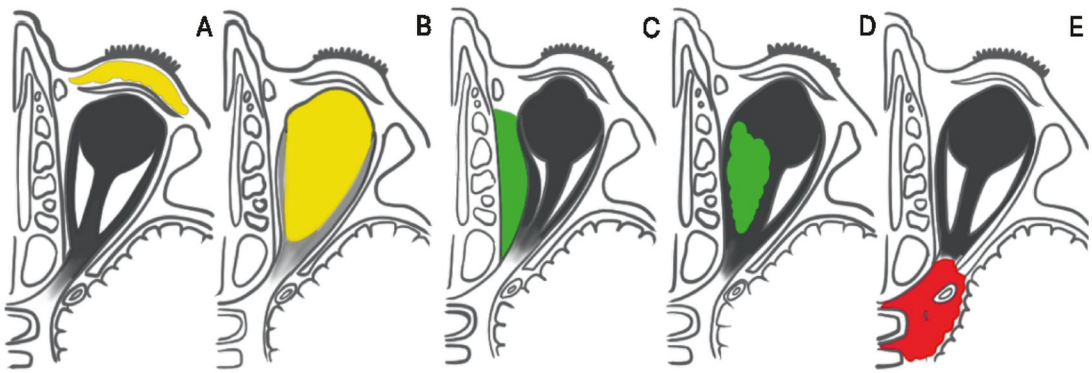




  - Osteomyelitis with swelling over frontal bone

Orbital Complications

- Preseptal cellulitis
- Orbital cellulitis
- Subperiosteal abscess
  - Pus accumulates between bone and periosteum
- Orbital abscess
  - Infection breaks periosteum and enters orbit
- Cavernous sinus thrombosis
  - Infection spreads from orbit via ophthalmic veins to cavernous sinus



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Intracranial Complications

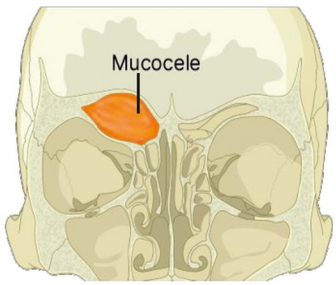
- Meningitis
- Extradural abscess
  - Pus between bone and dura
- Subdural abscess
  - Pus between dura and brain
- Brain abscess
  - Infection enters brain parenchyma

Descending Infections

- Nasal infection can spread to
  - Pharynx → pharyngitis
  - Larynx → laryngitis
  - Adenoids → adenoiditis
  - Tonsils → tonsillitis

MUCOCELE

00:19:50



- Most common involves Frontal sinus
- Least commonly involves sphenoid sinus
- Site: Superomedial quadrant of orbit
- Symptoms: Eye is displaced forward, downward, laterally
- Swelling: Cystic, non-tender, egg shell cracking
- X-ray: Loss of scalloped margins
- Treatment: Frontoethmoidectomy to promote drainage of frontal sinus

Intracranial Complications



- Extradural abscess
- Subdural abscess
- Meningitis
- Encephalitis
- Brain abscess
- Cavernous sinus thrombosis

CAVERNOUS SINUS THROMBOSIS

- Aetiology: Infection of ethmoids or sphenoid sinuses is most common cause
- But infection can spread from nose, ear, throat, orbit also
- Orbital cellulitis can secondarily lead to cavernous sinus thrombosis
- Treatment:
  - Immediate drainage of abscess and start IV antibiotics
  - Role of anticoagulants is not clear

Cavernous Sinus Thrombosis Vs Orbital Cellulitis

	Cavernous sinus thrombosis	Orbital cellulitis
Source	Nose, sinuses, orbit, ear or pharynx	Ethmoid sinuses

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Onset	Abrupt with chills and rigors	Slow
Toxemia	Present	Late
Laterality	Bilateral	Unilateral
Oedema of eyelids	Bilateral oedema of eyelids, chemosis and proptosis	Unilateral oedema of eyelids, near the inner canthus leading to chemosis and proptosis
Cranial nerve involvement	Involves individually and sequentially	Involved concurrently with complete ophthalmoplegia

Other Syndromes Similar To Cavernous Sinus Thrombosis

Condition	Cranial nerves involved	Cranial nerves not involved
Orbital apex syndrome	II, III, IV, VI ± V2	—
Superior orbital fissure syndrome	III, IV, VI ± V2	II
Cavernous sinus syndrome	III, IV, VI	II, V2

PYQ

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Q. An infant developed difficulty in breathing. His symptoms improve with crying. In which disease, do symptoms improve with crying? FMGE July 2023

A. Atrophic rhinitis
B. Acute rhinosinusitis
C. Allergic rhinitis
D. Choanal atresia

Answer: **D**

Q. During the discharge of a COVID patient who was treated with steroids and remdesivir, which of the following will you inform him about? INI CET July 2021

1. Repeat RT-PCR after 7 days of discharge
2. Watch for the persistence of Anosmia
3. Watch for headache and nasal discharge
4. Monitor glucose levels
5. Watch for Sinusitis like symptoms

A. 1, 3 and 4
B. 2, 3 and 4
C. 3, 4 and 5
D. 1,2,3,4 and 5

Answer: **C**

Q. A 30-year-old woman present with upper respiratory tract infection and orbital pain. What is the likely diagnosis?

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A. Atrophic rhinitis
B. Acute rhinosinusitis
C. Allergic rhinitis
D. Choanal atresia



Answer: **B**

Q. A patient presents with acute exacerbation of asthma; he gives a history of sinusitis and headaches for which he had self-medicated. Examination reveals nasal polyposis; which of the following drugs is likely implicated in this episode of asthma in this patient?

A. Cotrimoxazole
B. Co-amoxiclav
C. Aspirin
D. Chloramphenicol

Answer: **C**

Q. A 30-year-old patient with chronic sinusitis and nasal congestion, which anatomical structure is the most likely drainage pathway for the bulla ethmoidalis?

A. Middle meatus
B. Superior meatus
C. Inferior meatus
D. Middle concha

Answer: **A**

Q. Which of the following is not considered a major diagnostic criteria for allergic fungal sinusitis?

A. Fungal culture
B. Presence of Allergic Mucin
C. Characteristic CT findings: Double Density
D. Positive KOH stain

Answer: **A**

Q. Which of the following is true about orbital cellulitis?

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A. It is present anterior to the orbital septum
B. Treated effectively with topical antibiotics
C. Presents with proptosis, orbital swelling, normal pupil, and extraocular movements
D. Ethmoid sinusitis is the most common etiology

Answer: **D**